

1. Write short answers to the following questions:

- a. What does the word multimedia mean?
- b. Name the five main components of multimedia.
- c. Which element of multimedia is considered the most basic?
- d. Define graphics in multimedia.
- e. What does animation create?
- f. At what fps are movies usually shown?
- g. Give one advantage of multimedia.
- h. Mention one disadvantage of multimedia.
- i. What is the basic unit of raster graphics?
- j. Name one software used for editing graphics.
- k. Expand JPG.
- l. Which audio format provides uncompressed sound quality?
- m. Which video format is most widely used on YouTube?
- n. Give one use of computer graphics in the education sector.
- o. Name one software used for 3D Animation.
- p. Mention one example of 2D animation.
- q. What is resolution in video?
- r. Define a pixel.
- s. What does SVG stand for?
- t. Give one difference between raster and vector graphics.

2. Write the long answer to the following questions:

- a. Explain the concept of multimedia technology.
- b. What are the main advantages of multimedia?
- c. List three limitations of multimedia.
- d. Explain the role of text in multimedia.
- e. What are graphics and why are they important?
- f. Define animation with one example.
- g. What is video frame rate? Give examples.
- h. What is video resolution? Explain SD and HD.
- i. Differentiate between raster and vector graphics.
- j. Explain one application of computer graphics in medicine.
- k. Why is audio important in multimedia?
- l. Mention the difference between MP3 and AAC formats.
- m. What is Full HD and how is it different from 4K?
- n. Explain the use of MOV video format.
- o. What is tweening in 2D Animation?
- p. Explain the importance of 3D Animation in films.
- q. Define PDF and its main purpose.
- r. Mention two applications of computer graphics in simulation.
- s. Why is WAV not preferred for online sharing?
- t. Compare 2D and 3D animations with examples.

1) Short Answer Questions

a. What does the word multimedia mean?

Multimedia means the use of different forms of media such as text, images, audio, video, and animation together to present information.

b. Name the five main components of multimedia.

The five main components of multimedia are:

1. Text
2. Graphics/Images
3. Audio
4. Video
5. Animation

c. Which element of multimedia is considered the most basic?

Text is considered the most basic element of multimedia.

d. Define graphics in multimedia.

Graphics are visual images, drawings, charts, diagrams, or illustrations used to present information in multimedia.

e. What does animation create?

Animation creates the illusion of movement by displaying a series of images or frames in sequence.

f. At what fps are movies usually shown?

Movies are usually shown at approximately 24 frames per second (fps).

g. Give one advantage of multimedia.

Multimedia makes learning and communication more interesting and interactive.

h. Mention one disadvantage of multimedia.

Multimedia can require large storage space because it contains audio, video, images, and animations.

i. What is the basic unit of raster graphics?

The basic unit of raster graphics is a pixel (picture element).

j. Name one software used for editing graphics.

Adobe Photoshop is software used for editing graphics.

k. Expand JPG.

JPG stands for Joint Photographic Experts Group.

l. Which audio format provides uncompressed sound quality?

WAV (Waveform Audio File Format) provides uncompressed sound quality.

m. Which video format is most widely used on YouTube?

MP4 (MPEG-4) is one of the most widely used video formats on YouTube.

n. Give one use of computer graphics in the education sector.

Computer graphics are used to create educational diagrams, presentations, simulations, and animations.

o. Name one software used for 3D Animation.

Blender is software used for 3D animation.

p. Mention one example of 2D animation.

Tom and Jerry is an example of 2D animation.

q. What is resolution in video?

Video resolution is the number of pixels used to display a video, usually expressed as width $\times$ height, such as 1920×1080 .

r. Define a pixel.

A pixel is the smallest individual unit of a digital image or raster graphic.

s. What does SVG stand for?

SVG stands for Scalable Vector Graphics.

t. Give one difference between raster and vector graphics.

Raster graphics are made of pixels, while vector graphics are made using mathematical paths and shapes.

2) Long Questions – Simple Answers

a. Explain the concept of multimedia technology.

Multimedia technology is the use of different types of media together to present information. It includes **text, graphics, audio, video, and animation**.

It is used in education, entertainment, business, medicine, and communication. It makes information more **interesting, interactive, and easy to understand**.

b. What are the main advantages of multimedia?

The main advantages of multimedia are:

1. Makes learning easy and interesting.
2. Makes information attractive.
3. Provides interactive learning.
4. Improves communication.
5. Helps explain difficult topics using images, audio, and video.

c. List three limitations of multimedia.

Three main limitations of multimedia are:

1. It requires **large storage space**.
2. It can be **expensive** to develop.
3. It requires **technical knowledge** to create and operate.

d. Explain the role of text in multimedia.

Text is a basic element of multimedia. It is used to provide **information, titles, headings, labels, and instructions**.

Text also helps users understand images, videos, and animations and navigate through multimedia applications.

e. What are graphics and why are they important?

Graphics are **images, drawings, charts, diagrams, and illustrations** used in multimedia.

They are important because they make information **attractive, clear, and easy to understand**. Graphics are widely used in education, presentations, advertising, and entertainment.

f. Define animation with one example.

Animation is the process of creating the **illusion of movement** by showing a series of images or frames quickly.

Example: Cartoon characters moving in an animated movie.

g. What is video frame rate? Give examples.

Frame rate is the **number of frames displayed in one second of a video**. It is measured in **frames per second (fps)**.

Examples:

- Movies – **24 fps**
- Television/videos – **30 fps**
- Gaming – **60 fps**

h. What is video resolution? Explain SD and HD.

Video resolution is the **number of pixels used to display a video**.

- **SD (Standard Definition)**: Has lower resolution and less detail.
- **HD (High Definition)**: Has higher resolution and better picture quality, such as **1280 × 720** or **1920 × 1080**.

i. Differentiate between raster and vector graphics.

Raster Graphics	Vector Graphics
Made of pixels	Made of lines and shapes
Good for photographs	Good for logos and drawings
May lose quality when enlarged	Does not lose quality when enlarged
Example: JPG, PNG	Example: SVG

j. Explain one application of computer graphics in medicine.

Computer graphics are used in medicine to create and display **medical images and 3D models**.

They help doctors examine internal body parts, detect diseases, and plan treatments or surgeries. **CT scans and MRI images** are examples.

k. Why is audio important in multimedia?

Audio adds **voice, music, and sound effects** to multimedia.

It makes videos, animations, and presentations more **interesting and realistic**.

Audio is also useful for narration and educational content.

l. Mention the difference between MP3 and AAC formats.

MP3 and AAC are both **compressed audio formats**.

- **MP3**: Very popular and widely supported.
- **AAC**: Generally provides better sound quality at a similar bit rate.

Both formats are commonly used for digital audio.

m. What is Full HD and how is it different from 4K?

Full HD has a resolution of 1920×1080 pixels.

4K commonly has a resolution of 3840×2160 pixels.

Therefore, 4K has more pixels and provides **more detail and sharper images** than Full HD.

n. Explain the use of MOV video format.

MOV is a **video file format developed by Apple**. It is used to store video and audio in one file.

It is commonly used for **video editing, storing high-quality videos, and multimedia production**. Its file extension is **.mov**.

o. What is tweening in 2D Animation?

Tweening is the process of creating **intermediate frames between two keyframes**.

For example, if an object moves from the left side to the right side, tweening creates the frames between these two positions. It helps produce **smooth animation**.

p. Explain the importance of 3D Animation in films.

3D animation is used to create **three-dimensional characters, objects, and environments**.

It is important in films because it helps create **realistic characters, special effects, imaginary worlds, and attractive scenes**. It is commonly used in animated and science-fiction movies.

q. Define PDF and its main purpose.

PDF stands for **Portable Document Format**. It is a file format used to store and share digital documents.

Its main purpose is to **preserve the original formatting and layout** of a document when it is opened on different devices.

r. Mention two applications of computer graphics in simulation.

Two applications of computer graphics in simulation are:

1. **Flight simulation:** Used for pilot training.
2. **Medical simulation:** Used to practice medical procedures.

Computer graphics make these simulations more realistic and interactive.

s. Why is WAV not preferred for online sharing?

WAV usually stores **uncompressed audio**, so its file size is large.

Large WAV files require more **storage space, internet bandwidth, and downloading time**. Therefore, compressed formats such as MP3 are usually preferred for online sharing.

t. Compare 2D and 3D animations with examples.

2D Animation	3D Animation
Looks flat	Has depth and looks more realistic
Uses height and width	Uses height, width, and depth
Generally simpler	Generally more complex
Example: 2D cartoons	Example: 3D animated movies

2D animation is commonly used for **cartoons and simple illustrations**, while 3D animation is used for **films, games, and realistic visual effects**.

3. Write the technical term for the given statements

No.	Statement	Answer
a.	A format used for animated web graphics.	GIF
b.	A text + image + audio + video integration system.	Multimedia
c.	The smallest unit of a digital image.	Pixel
d.	File format that supports transparency and compression.	PNG
e.	A method of creating motion by rapidly displaying still frames.	Animation
f.	A file type for scalable web graphics based on XML.	SVG
g.	High-quality uncompressed audio format.	WAV
h.	A video format developed by Apple for QuickTime.	MOV
i.	A format for sharing reports and e-books across devices.	PDF
j.	A free lossless audio codec used for archiving.	FLAC
k.	A raster image format best for photographs.	JPG / JPEG
l.	A software used for 3D modeling and animation.	Blender

No.	Statement	Answer
m.	A format once widely used for Flash video streaming.	FLV
n.	The speed of video frames shown per second.	Frame Rate (FPS)
o.	A widely used compressed audio format.	MP3
p.	A type of graphics that uses mathematical equations.	Vector Graphics
q.	A multimedia application for medical scanning images.	DICOM
r.	An early Disney movie that used 2D animation.	Snow White and the Seven Dwarfs
s.	A 3D animated movie that marked a new era in film industry.	Toy Story
t.	A type of animation using flat surfaces with only height and width.	2D Animation

4. Full Forms

No.	Short Form	Full Form
a.	JPG	Joint Photographic Experts Group
b.	GIF	Graphics Interchange Format
c.	PNG	Portable Network Graphics
d.	SVG	Scalable Vector Graphics
e.	PDF	Portable Document Format
f.	MP3	MPEG-1 Audio Layer III
g.	AAC	Advanced Audio Coding
h.	WAV	Waveform Audio File Format
i.	FLAC	Free Lossless Audio Codec
j.	ALAC	Apple Lossless Audio Codec
k.	CAD	Computer-Aided Design
l.	AVI	Audio Video Interleave
m.	MKV	Matroska Video
n.	FLV	Flash Video
o.	WebM	Web Media
p.	WMA	Windows Media Audio

No.	Short Form	Full Form
q.	MIDI	Musical Instrument Digital Interface
r.	GUI	Graphical User Interface
s.	MPEG	Moving Picture Experts Group

5. Multiple Choice Questions

1. Multimedia is best defined as:

b. Integration of multiple media elements in digital form

2. One limitation of multimedia content is that it:

a. Requires a large storage space

3. Which software is commonly used for editing graphics?

b. Adobe Photoshop

4. A video frame is:

b. A single image in a video sequence

5. In raster graphics, image quality depends on:

a. Number of pixels

6. Which of the following is a vector file format?

d. SVG

7. The main advantage of vector graphics is that they:

b. Maintain quality at any size

8. GIF format is especially suitable for:

b. Animated web graphics

9. PNG format is preferred when:

a. Transparency is required

10. FLAC is an example of:

c. Lossless audio format

11. AAC is commonly used in:

a. Streaming platforms

12. 480p resolution is categorized as:

a. SD

13. 4K resolution offers:

c. Four times the detail of Full HD

14. MOV format is mainly popular among:

a. Professional video editors

15. The AVI format was developed by:

d. Microsoft

16. WebM format is mainly designed for:

c. Modern web browsers